- *Simplify the data collection process*. Reduce the friction taken for manual data collection by providing an interface that is easy to use and is available in the field where the measurement is taken. Try to eliminate the need to transcribe data by providing a mobile interface where it can be entered directly as the measurement is taken rather than writing it down and then having to input it when back in the office.
- *Provide immediate value*. Providing feedback during the data entry, through alerts when readings are out of normal ranges, or showing trends makes the benefits of performing the data entry tangible and motivates users to continue with the process.
- *Automate reminders*. Consider providing useful automated reminders that prompt users to perform manual collection. If you can make these smarter than just a daily reminder—for example by taking into consideration context such as the amount of time since the last test, or abnormal weather that may impact levels, all the better.
- *Design for scale from the start*. When dealing with hundreds of collectors taking thousands of readings, the challenges multiply. Build in quality controls such as validation rules, duplicate detection, and automated data quality checks to catch errors early. Consider implementing role-based access and workflows that allow supervisors to review and approve submissions before they enter the main dataset. Use dashboards that give managers visibility into collection completion rates across teams, identifying gaps or patterns that need attention.

3.4.4 Integrating third-party data

Many physical parameters affecting your system are already being measured by others. Weather services track temperature, humidity, rainfall, and solar radiation while utilities monitor grid conditions and traffic services track road conditions. The challenge can be accessing and integrating this data. In my digital twin I integrate with an external weather data source to obtain forecast data, and with the water corporation to obtain data about water restrictions, as well as the state of the metropolitan water supplies.